School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Project Scribe Submission

Adaptive Traffic Control Using Las Vegas Algorithm

Group Number: S1_G7_ITS

Date: March 10$^{\text{th}}$, 2026

Scribe Question 1: System Objective and Deterministic Baseline

Clearly restate your project objective.

Identify and describe the deterministic baseline strategy you are implementing. Explain why this baseline is reasonable or commonly used in your domain.

Answer:

Project Objective

this project aims at designing and simulating an adaptive traffic signal controller system at a signalized intersection through real-time traffic data on SUMO through TraCI. The system aims at using dynamically changing traffic signal timings to enhance traffic movement and minimize congestion.

Specifically, the system aims to minimize:

- Average vehicle waiting time
- Average travel time
- Queue length at intersections

While improving:

- Average vehicle speed
- Overall traffic throughput.

The controller also observes the real-time traffic parameters like the number of vehicles waiting in a queue, the rate of passage of vehicles and their turning direction and employs this to calculate the best timing of the traffic lights.

Deterministic Baseline Strategy

The deterministic baseline adopted in this project is Webster Cycle Length Optimization technique, and proportional green time to be used in the phases.

The Webster formula is used to find out the best cycle length of the traffic signal depending on the demand in traffic.

The optimal cycle length is computed as:

$$C_o = \frac{1.5L + 5}{1 - \sum y_i}$$

Where,
C_o = optimal cycle length
L = total lost time per cycle
y_i = critical flow ratio for phase i

The critical lane flow ratio is defined as:

$$y_i = \frac{max(\varepsilon_s, \varepsilon_l, \varepsilon_{sr})}{s}$$

Where,
ε_s = straight movement flow
ε_l = left-turn flow
ε_{sr} = right-turn flow
s = saturation flow rate (vehicles/hour)

After computing the cycle length, the green time is distributed proportionally across phases according to their traffic demand.

$$g_i = \frac{y_i}{\sum y_i}(C_o - L)$$

Where,
g_i = green time for phase i

Why This Baseline is Reasonable
Webster method is often applied in traffic engineering and signal timing design as it offers a mathematically motivated approach of allocating green times depending on the traffic demand.

The positive aspects of this baseline are:

- Reckless deterministic decision rule.
- Effective at crossroads.
- Simulation environments (such as SUMO) are easy to implement.

The controller in this system determines the timing of signal execution deterministically in response to measured traffic conditions, and guarantees that results are reproducible between simulation runs.

Scribe Question 2: Random Variables in the Simulation
Identify the key random variables in your system. For each variable:

- Provide a formal definition.
- Specify the assumed distribution.
- Explain how it is used within the deterministic baseline simulation.

Note: The decision rule must be deterministic, even if uncertainty exists in the system inputs

Answer: Our system consists of the following key random variables:

- **Vehicle Arrivals (N_t):**

  Definition
  The traffic flows at the intersection are represented as a stochastic process whereby vehicles can join the road system at any time randomly.
  let, $A(t)$ represent the number of vehicles arriving during time interval t .

  Assumed Distribution
  Poisson distribution is commonly applied in vehicle arrival models and is mostly applied in traffic flow models.

  $$P(A = K) = \frac{\lambda^k e^{-\lambda}}{k!}$$

  Where,
  k = number of arriving vehicles
  λ = average arrival rate

  Use in Simulation
  The arrival of vehicles in the SUMO simulation is done by creating trip files and route definitions. Such arrivals cause random demand in the traffic which the deterministic controller is required to control.

- **Turning Movement Distribution :**

  Definition
  Vehicle arriving at intersection may choose different direction like straight, right, left.
  Let,

  $$T \in \{s, l, r\}$$

  represents the turning of vehicle

  Assumed Distribution
  It is modeled using categorical probability distribution:

  $$P(T = s) = p_s, P(T = l) = p_l, P(T = r) = p_r$$

  Where,

  $$p_s + p_l + p_r = 1$$

  Use in Simulation
  These turning movements affect the traffic demand of the various phases of the signal and is calculated to determine the critical lane flow ratio y_i.

- **Vehicle Speeds (V):**

  Definition
  vehicle speed varies based on traffic density and driver behavior. Let, V_i represents the speed vehicle i.

  Assumed Distribution
  vehicle speed can be modeled using Normal distribution:

  $$V_i \sim N(\mu, \sigma^2)$$

  Where,
  μ = Mean traffic speed
  σ^2 = Speed variance

  Use in Simulation
  Speed data is collected from SUMO using traci.vehicle.getSpeed() and it is used to compute performance metrics such as average network speed.

Scribe Question 3: Probabilistic Modeling and Fixed Decision Rule
Explain how uncertainty is modeled in your simulation while keeping the decision strategy deterministic.

- What probabilistic assumptions are being used?
- How does the deterministic baseline operate under uncertain inputs?
- Provide relevant mathematical expressions if applicable.

Answer:

Modeling Uncertainty
Even though the traffic signal controller operates using a deterministic decision rule, the traffic environment is uncertain by nature as it can be affected by the arrival of different vehicles, driver behaviors, and routes taken by vehicles.

The simulation reflects these uncertainties by means of stochastic processes including:

- Random vehicle arrival times
- Random turning movements
- Variation in vehicle speeds

These uncertain patterns of traffic are created in the SUMO simulation and the controller views the consequent traffic situation in real time.

Deterministic Decision Rule
The signal control strategy is deterministic in spite of the stochastic inputs.

The controller calculates the optimum cycle length at the completion of every cycle by:

$$C_o = \frac{1.5L + 5}{1 - \sum y_i}$$

The critical flow ratios y are calculated based on the counts of vehicles observed.
After calculating the cycle length, green times will be allocated as:

$$g_i = \frac{y_i}{\sum y_i}(C_o - L)$$

These equations are a mathematical rule that is fixed and which transforms traffic observations into signal timings.

As such, the controller will give the same output with a given set of inputs.

Operation Under Uncertain Inputs
Although the arrival of vehicles as well as the movement is random, the controller responds deterministically by computing the signal timings according to the current measurements of the traffic.

This enables the system to react to the changes in traffic demand without altering a decision rule.

End of Submission